Final Exam - Sp (2022)

Ans-to the ques no - 1(a)

28/22

• Define: i) Lattice. ii) Basis. iii) Crystal. iv) Unit Cell. v) Crystalline solid.

⇒

[AU-18] **Lattice:** The regular periodic arrangements of atoms, ions and molecules in space which looks like a net is called lattice.

Basis: The structure of all crystal can be described in terms of a lattice, with a group of atoms attached to every lattice point. The group of atoms is called basis. it is replaced in space to form ~~of~~ the crystal structure.

Crystal: A crystal is a regular array of three dimensional units. The structural units are atoms, ions and molecules or group of atom. An ideal crystal is constructed by the infinite repetition of identical structural unit is space. In the simplest crystal the structural unit is a sigle atom, as in copper, silver, gold, iron, aluminium.

Unit cell: The lattice array of a group is a periodic repetition of a fundamental grouping. Therefore it is not necessary to consider the entire array. This is done by selecting a region of the crystal defined by three vectors $\vec{a}, \vec{b}, \vec{c}$ translated by any integral multiple of these vectors, reproduce a similar regim of the crystal. The regim is called a unit cell and these vectors as basis vectors.

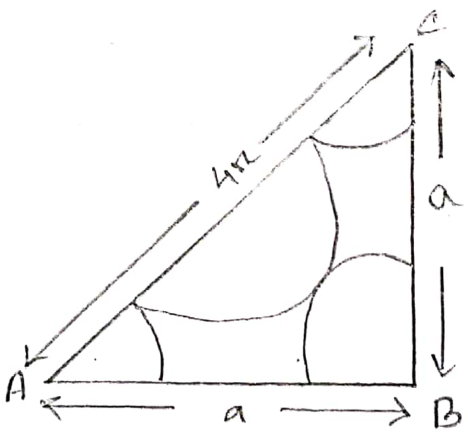
Crystalline solid: In crystalline solid, the atoms are stacked in a regular ~~mann~~ manner in crystals.

In this types of crystals, the atomic array is periodic that is a representative unit or motif is repeated at regular intervals along any and all directions in the crystal.

Ans to the ques no - 1 (b)

• Show that the packing function for a fcc lattice is 0.74 and hence write the meaning of this ~~members~~. number.

In the unit cell of fcc lattice there are eight atoms per unit cell suitable at the eight corners and one atom at the centre of each face.



$AC = r + 2r + r$

$= 4r$

$AC^2 = a^2 + a^2$ 3

$\Rightarrow (4r)^2 = 2a^2$

$\Rightarrow 16r^2 = 2a^2$

$\Rightarrow a^2 = \frac{\cancel{16}^{8} r^2}{\cancel{2}}$

$\Rightarrow a^2 = 8r^2$

$\Rightarrow \sqrt{a^2} = \sqrt{8r^2}$

$\therefore a = 2\sqrt{2}r$

Total volume of the unit cell $= a^3$

$= (2\sqrt{2}r)^3$

$= 16\sqrt{2}r^3$

atom per unit cell $= 4$

$\therefore$ The volume of the atoms per unit cell $= 4 \times \frac{4}{3}\pi r^3$

$= \frac{16}{3}\pi r^3$

$\therefore P.F = \frac{\frac{16}{3}\pi r^3}{\frac{16\sqrt{2}}{1} r^3}$

$= \frac{\cancel{16}}{3}\pi \cancel{r^3} \times \frac{1}{\cancel{16}\sqrt{2}\,\cancel{r^3}}$

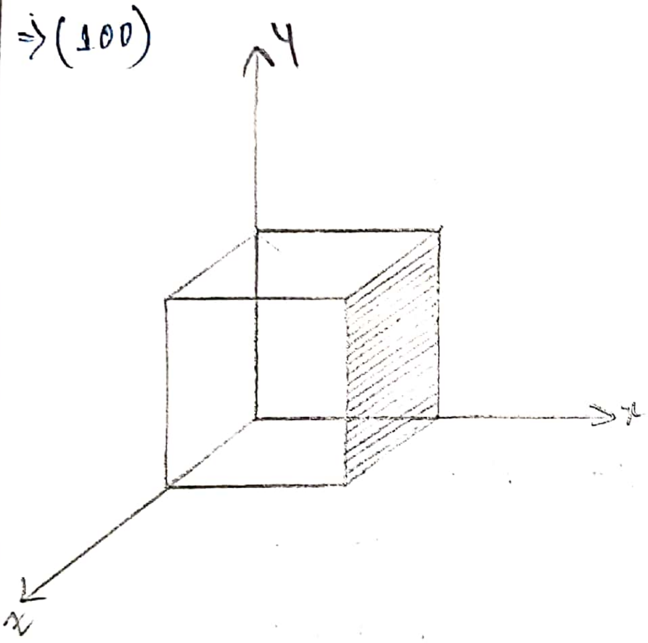
$= \frac{1}{3\sqrt{2}}\pi$

$= 0.74.$

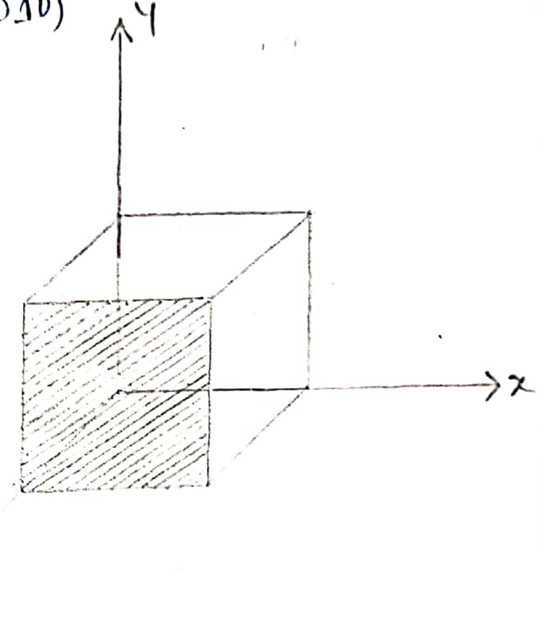
Ans-to-the ques no-2(a)

• Define Miller indices and draw the plane of (100), (010), (101). and (111)

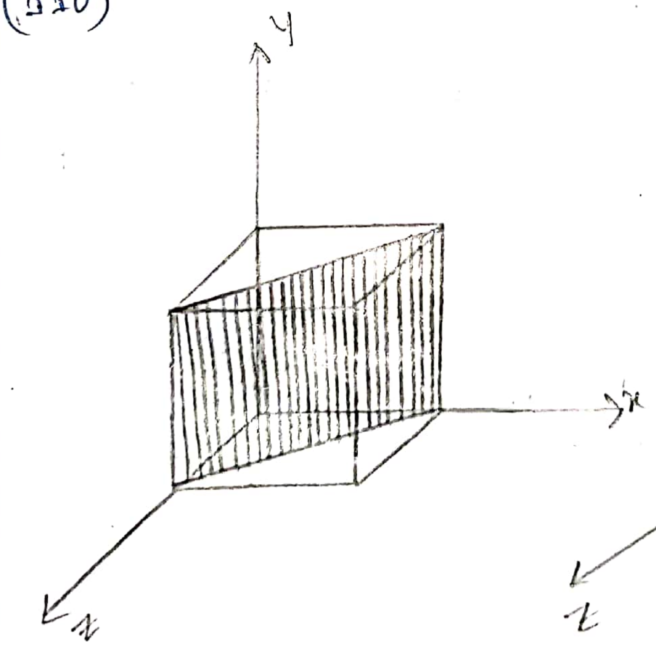
⇒ (100)



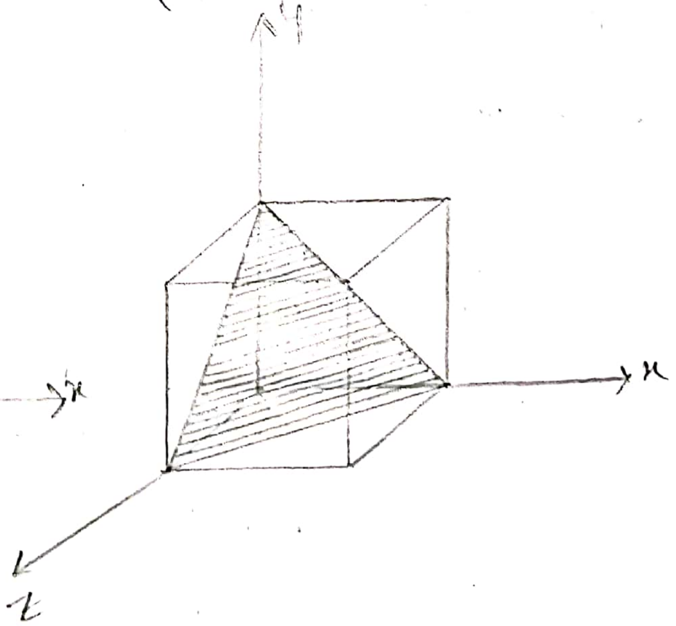
(010)



(110)

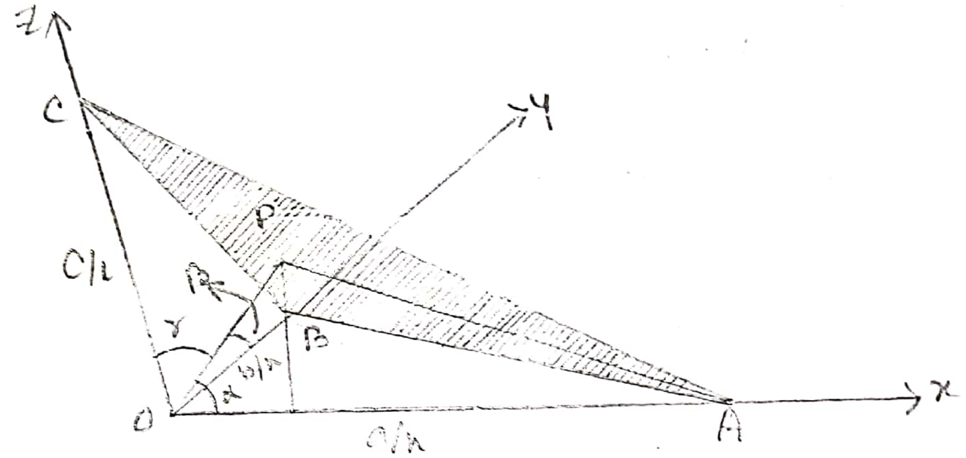


(111)



Ans to the ques no - 2(b)

[AU-18] • Show that in a crystal of cubic structure, the distance between the planes with Miller indices h, k, l is equal to $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$, where a is the lattice parameter.



$OA = a/h$

$OB = b/h$

$OC = c/l$

$$\cos\alpha = \frac{d}{OA} = \frac{d}{a/h} = \frac{dh}{a}$$

$$\cos\beta = \frac{d}{OB} = \frac{d}{b/h} = \frac{dh}{b}$$

$$\cos\gamma = \frac{d}{OC} = \frac{d}{cl} = \frac{dl}{c}$$

Cosine direction, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$$\left(\frac{dh}{a}\right)^2 + \left(\frac{dh}{b}\right)^2 + \left(\frac{dl}{c}\right)^2 = 1$$

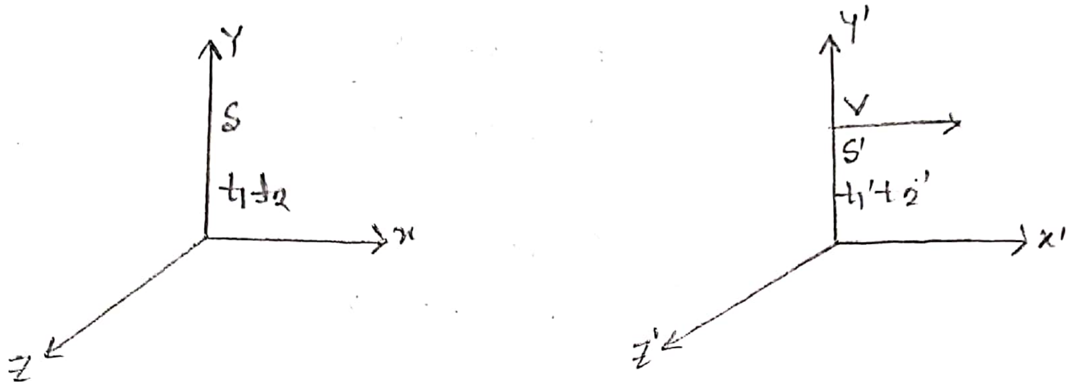
For cubic lattice,

$$a = b = c$$

$$d = \frac{a}{\sqrt{h^2+k^2+l^2}}$$

Ans to the ques no - 3(a)

• Explain " A moving clock always appers to go slow".



Let us consider two frames of reference S and S'. S is moving with velocity belong to +ve dimention of x-axis relative S.

Let, a ~~color~~ clock be placed in reference system we have,

$$t_1' = \frac{t_1 - \frac{Vx}{c^2}}{\sqrt{1-\frac{V^2}{c^2}}} \quad \text{——— (1)}$$

If this clock again gives a signal at time t_2,

$$t_2' = \frac{t_2 - \frac{Vx}{c^2}}{\sqrt{1-\frac{V^2}{c^2}}} \quad \text{——— (2)}$$

When this intercval as measured from the moving system it is equal to $t_2' - t_1' = \Delta t$

Thus from equations (1) & (2) we have,

$$\Delta t' = t_2' - t_1'$$

$$\Rightarrow \Delta t' = \frac{t_2 - t_1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow \Delta t' = \frac{\Delta t}{\sqrt{1 - \frac{v^2}{c^2}}}$$

i.e,

$$\Delta t' > \Delta t$$

Now, if we consider the clock placed in S: then clock is at rest relative to S' and is moving the velocity v relative to S.

We get,

$$t_1 = \frac{t_1' + \frac{Vx}{Vc}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{———} \quad (3)$$

$$t_2 = \frac{t_2' + \frac{Vx}{Vc}}{\sqrt{1 - v^2/c^2}} \quad \text{———} \quad (4)$$

From eq^n (3) & (4)

$$t_2 - t_1 = \Delta t = \frac{t_2 - t_1}{\sqrt{1 - V^2/c^2}}$$

$$\Rightarrow \Delta t = t_2 - t_1 = \frac{\Delta t}{\sqrt{1 - V^2/c^2}}$$

i.e, $\Delta t > \Delta t'$

This from above reasoning we may say, "A moving ~~check~~ clock always appears to go slow".

Ans to the ques no-4(a)

• State the laws of radioactive ~~dist~~ disintegration. Show that the number of radioactive atoms decreases exponentially with time.

Law of radioactive Disintegration: Radioactive disintegration is found to obey the following laws:

1. Atoms of all radioactive elements undergo spontaneous disintegration to form fresh radioactive products with the emission of α-, β-, γ- rays.

2. The rate of radioactive disintegration i.e, the number of disintegration per second is not affected by environmental factors (Like temperature, pressure and chemical

Combinatim etc) but depends on the number of the atoms of the original kind present at any time.

Here,

N_0 = The number of radioactive present in a sample at the beginning of disintegration

$t = 0$

N = at anytime t

dN = at a time dt.

$$\frac{dN}{dt} \propto -N$$

$$\cancel{dN} \Rightarrow \frac{dN}{dz} = -\lambda N$$

$$\Rightarrow \frac{dN}{N} = -\lambda dt$$

$$\Rightarrow \int \frac{dN}{N} = -\lambda \int dt$$

$$\Rightarrow \ln N = -\lambda t + C \quad \text{--- (i)}$$

When $t = 0$, $N = N_0$

$\ln N_0 = C$

Substituting this value of C in eqn (i)

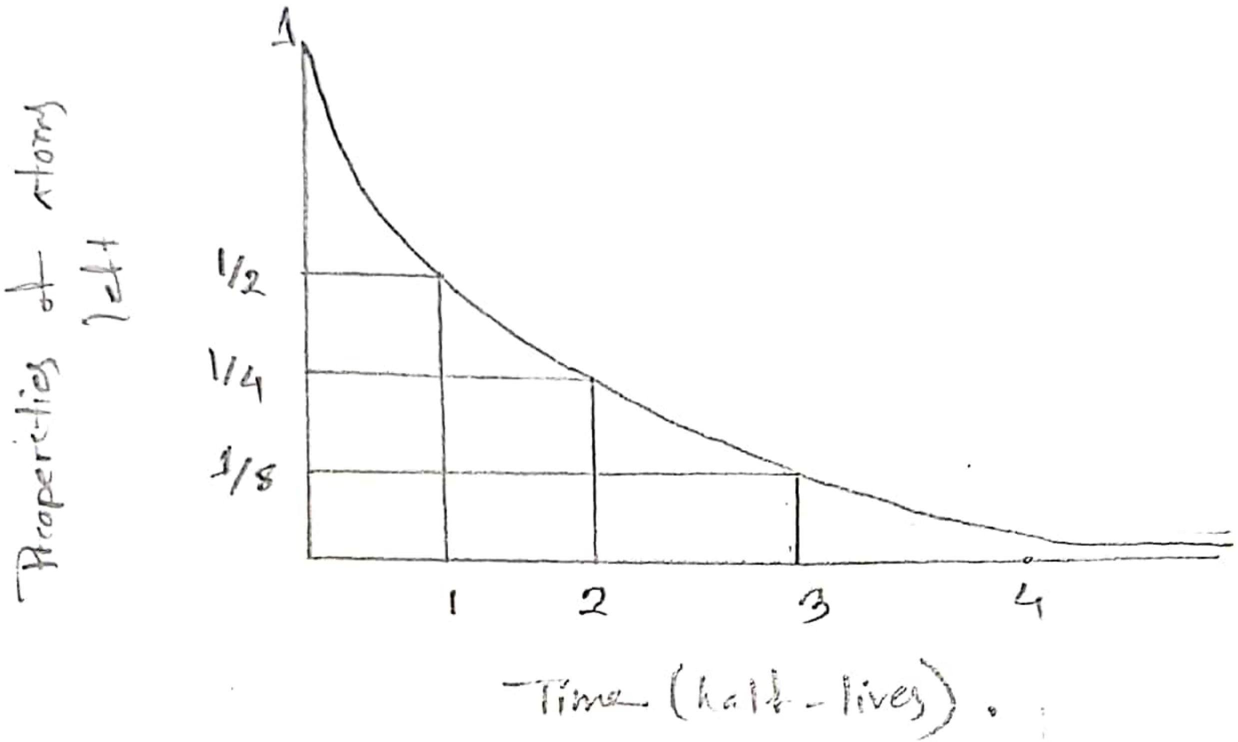
$$\ln N = -\lambda t + \ln N_0$$

$$\Rightarrow \ln N/N_0 = -\lambda t$$

$$\Rightarrow N/N_0 = e^{-\lambda t}$$

$$\therefore N = N_0 e^{-\lambda t}$$

Show that the number of radioactivity atom decreases expotentially with time.



Ans - To the ques no - 4(b)

• The half life of a radioactive substance in 30 days Calculate i) the radioactive decay constant,
ii) the mean life , iii) the time taken for (1/8) for the original number of atoms to remain unchanged.

(i)

The radioactive decay constant —

$T_{1/2} = 30$ days

$\therefore \lambda = \frac{0.693}{30}$

$= 0.0231$ per day.

(ii)

The mean life, $\tau = \frac{1}{\lambda}$

$= \frac{1}{0.0231}$

$= 43.29$ days.

Here,

$\lambda = 0.0231 \, d^{-1}$

(iii)

The number of atoms left, $N = \frac{1}{8} N_0$

$\therefore \frac{N}{N_0} = \frac{1}{8} = e^{-\lambda t}$

or; $e^{-\lambda t} = 8$

$\Rightarrow \lambda t = \log_e 8$

$\Rightarrow t = \frac{\log_e 8}{\lambda}$

$\Rightarrow t = \frac{\log_e 8}{0.0231}$

$\therefore t = 90$ Days.

Ans to the ques no - 5(a)

•Define 'Nuclear Fission' and 'Nuclear Fusion' with an example of each.

Nuclear Fission: The process of breaking up of the nucleus of a heavy atom into two more or less equal fragments with the release of a large amount of energy is known as fission.

$${}_{92}U^{235} + {}_0n^1 \rightarrow {}_{92}U^{236} \rightarrow X + Y + \text{neutrons} \quad \text{——— (1)}$$

$${}_{92}U^{235} + {}_0n^1 \rightarrow {}_{92}U^{236} \rightarrow {}_{56}Ba^{141} + {}_{36}Kr^{92} + 3\,{}_0n^1 + Q \quad \text{——— (2)}$$

$${}_{92}U^{235} + {}_0n^1 \rightarrow {}_{92}U^{236} \rightarrow {}_{56}Xe^{140} + {}_{38}Sr^{94} + 2\,{}_0n^1 + Q \quad \text{——— (3)}$$

Nuclear fusion: The process in which two or more light nuclei combined into a single nucleus with the release of tremendous amout of every is called as nuclear fussion.

Ans to the ques no-5(b)

• Find the nuclear Mass, Size, density, binding energy and binding energy and binding energy per nucleon for $_{88}Ra^{226}$.

⇒ We know that

$M_p = 1.007276$ a.m.u

$M_n = 1.008665$ a.m.u

1 a.m.u $= 1.66 \times 10^{-27}$

$c^2 = 931$ MeV

Atomic mass nucleas $= (88 \times 1.007276 + 138 \times 1.008665)$

$= (88.640288 + 139.19577)$

$= 227.836058$

$= (227.836058 \times 1.66 \times 10^{-27})$ kg

$= 3.78 \times 10^{-25}$ kg

nucleas Size, $R = r_0 A^{1/3}$ $[r_0 = 1.3 \times 10^{-15}]$

$= (1.3 \times 10^{-15} \times 226)^{1/3}$

$= (2.9 \times 10^{-13})^{0.33}$

$= \cancel{27.3 \times 10^{-5}}$

$= 9.7 \times 10^{-14}$

$$\text{Nuclear density} = \frac{\text{Nuclear mass}}{\text{Nuclear Volume}}$$

$$= \frac{Zm_p + Nm_n}{4/3\,\pi R^3}$$

$$= \frac{3.78\times10^{-25}}{\frac{4}{3}\times3.1416\times(9.7\times10^{-14})^3}$$

$$= \frac{3.78\times10^{-25}}{3.82\times10^{-39}}$$

$$= 9.8\times10^{13}\ \text{kgm}^{-3}$$

$$\text{Nuclear charge} = Ze$$

$$= 88\times1.6\times10^{-19}$$

$$= 1.4\times10^{-17}\ \text{C}$$

$$\text{Mass defect} = M - A$$

$$= 227.836058 - 226$$

$$= 1.836058\ \text{a.m.u}$$

$$\text{Nuclear binding energy, B.E} = \{(Zm_p + Nm_n) - A\}\times c^2$$

$$= (227.836058 - 226)\times931$$

$$= 1709.36\ \text{MeV}$$

$$\text{Binding energy per nucleon} = \frac{\text{Binding Energy}}{A}$$

$$= \frac{1709.36}{226}$$

$$= 7.56\ \text{a.m.u}$$